Algorithmic Audit & Awareness Campaign

Social Media News Feed Algorithm

Faheem Abdul • Purnachandra Rao Birkur • Pranith Kasireddy
Abdul Rahbar Mohammed • Harish Pasupuleti







What Are Algorithms?

Definition

A set of step-by-step instructions that a computer follows to solve a problem or make a decision, processing vast amounts of data in milliseconds.

In Your Everyday Life

Google Search
Ranks billions of pages

Amazon
Recommends products

Spotify
Curates your playlists

News Feeds
Filters what you see

Algorithms make over **35,000 decisions** per second on platforms like Facebook

Our Focus: Social Media News Feed

2B+

Instagram Monthly Active Users

3B+

Facebook Monthly Active Users

2.5 hrs

Average Daily Social Media Time

Why This Matters

Unlike search engines where you actively seek information, social media feeds are passively consumed. The algorithm decides what you see without you asking — curating, filtering, and prioritizing content to keep you engaged the longest, not to inform you best.

How the News Feed Algorithm Works

1



Data Collection

Gathers signals from likes, shares, watch time, and scrolling behavior

2



Content Ranking

Assigns relevance scores based on hundreds of engagement factors

3



Engagement Prediction

Estimates the probability you will interact with each post

4



Personalized Delivery

Delivers highest-scoring content at the top of your feed

Key Insight: The primary goal is to maximize your time on the platform — not to inform or educate you.

The Data Behind Your Feed



Behavioral Data

Every tap, swipe, like, share, comment, and pause is tracked



Demographic Data

Age, gender, location, language, and education level



Social Graph

Your connections, interactions, and group memberships



Content Analysis

Text, images, hashtags, and video analyzed by AI models



The Core Concern

Training data reflects existing societal patterns and prejudices.

If historical data contains biases — racial, gender, political, socioeconomic — the algorithm learns and amplifies those biases at massive scale.

Types of Algorithmic Bias



Filter Bubble

Shows content similar to past engagement, creating an information bubble with only confirming viewpoints.

You think you see a balanced view — you're seeing a curated echo chamber.



Engagement Bias

Prioritizes content generating strong emotional reactions — outrage, fear, anger — over balanced information.

Sensational content is boosted; thoughtful, moderate content gets buried.



Representation Bias

Training data underrepresents certain demographics, causing unequal visibility and higher suppression rates.

Minority voices are suppressed while mainstream content is amplified.

Real-World Impact of Algorithmic Bias

64%

Political Radicalization

of people who joined extremist groups on Facebook were led there by the recommendation algorithm

6×

Misinformation Spread

False news spreads six times faster than true news on social media (MIT, 2018)

Areas of Concern



Mental Health

Instagram's own research found its algorithm worsens body image issues, especially among teenage girls



Political Polarization

Filter bubbles create separate realities where opposing sides see completely different information



Information Quality

Algorithmic amplification of sensational content degrades the quality of public discourse

Sources: MIT Media Lab (2018), Wall Street Journal — Facebook Files (2021), Facebook Internal Research

Our Algorithmic Audit: Key Findings

Methodology: *Two-week analysis of group members' social media feeds, combined with academic research review*

01 Content Homogeneity

After 3 days of engaging with one topic, over 70% of feed content related to that single topic

02 Emotional Amplification

Posts with negative or outrage-inducing content appeared 3x more frequently than neutral content

03 Demographic Imbalance

Content creators from certain backgrounds received disproportionately higher algorithmic visibility

04 Lack of Transparency

No platform explanation for why content was shown, and no meaningful user control over ranking

Our Awareness Campaign Strategy



EDUCATE

- Break down how algorithms work in simple terms
- Visual explainers for non-technical audiences
- Interactive demos showing feed personalization



EXPOSE

- Real examples of different feeds for different users
- Data-driven infographics on bias patterns
- Case studies of documented algorithmic harm



EMPOWER

- Actionable steps to reduce algorithmic influence
- Tool recommendations for feed control
- Challenge: go chronological for one week

Target Audience: University students, delivered through Instagram Stories, short-form video, and shareable infographics

Our Digital Campaign Artifacts



Infographic

"What Your Feed Isn't Showing You"

Visual map of how the algorithm filters content and what gets hidden. Designed for instant sharing on Instagram and WhatsApp.

Instagram Story



Instagram Carousel

"Break Your Bubble"

Swipeable cards teaching 5 steps to diversify your feed. Each card features one clear action with a simple visual.

5-Slide Carousel



Social Challenge

"Algorithm Audit Challenge"

Users switch to chronological feed for one week and document changes they notice. Participatory format for firsthand experience.

Reels + Hashtag

Be a Critical Consumer

Five actions you can take today

Diversify Your Sources Follow accounts with different perspectives — step outside your comfort zone

Check Your Settings Most platforms offer chronological feed options — use them

Engage Mindfully Every like and share trains the algorithm — be intentional

Verify Before Sharing Check accuracy first — bias means sensational content reaches you first

Spread Awareness Share what you've learned — understanding weakens biased systems



Your feed is not reality.

It's a curated selection designed to keep you engaged.

Question it.

Thank you for your attention — Questions?

References

1. Pariser, E. (2011). The Filter Bubble: How the New Personalized Web Is Changing What We Read and How We Think. Penguin Books.

2. Noble, S. U. (2018). Algorithms of Oppression: How Search Engines Reinforce Racism. NYU Press.

3. O'Neil, C. (2016). Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy. Crown.

4. Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. Science, 359(6380), 1146–1151.

5. Horwitz, J. & Seetharaman, D. (2021). Facebook Knows Instagram Is Toxic for Teen Girls, Company Documents Show. The Wall Street Journal.

6. Pew Research Center (2024). Social Media Use in 2024. Pew Research Center.

7. Meta Transparency Center (2023). How Feed Works. Meta Platforms, Inc.

8. Bakshy, E., Messing, S., & Adamic, L. A. (2015). Exposure to ideologically diverse news and opinion on Facebook. Science, 348(6239), 1130–1132.